To what extent does listening to music lessen the symptoms of ADHD

Psychology Extended Essay

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Introduction

About 2-5% of children in the UK are considered to have attention deficit, hyperactivity disorder (ADHD) in the UK (OptimistMinds, n.d.). There is also a higher rate of diagnosis in boys than in girls due to symptoms in girls often presenting themselves differently, sometimes leading to underdiagnoses (PsychCentral, n.d.). The average age of diagnoses in the UK is between the ages of 3-7 years old (NHS, n.d.).

The most commonly agreed on symptoms of ADHD are said to be difficulty paying attention and hyperactivity, however, further research has provided evidence that symptoms can go beyond this, taking the form of anxiety, depression, sleep problems and poor organisational skills (NHS, n.d.). According to the DSM-5 diagnostic criteria, for an ADHD diagnosis in an adult, there must be 5 or more symptoms of inattention and/or hyperactivity, some of these symptoms were present before the age of 12, symptoms must be present in 2 or more situations such as at home, school or work, there is evidence that symptoms interfere and disrupt every day life, making aspects such as social or academic situations more difficult and finally making sure the symptoms are not present with schizophrenia or cannot be explained by other disorders such as mood disorders of anxiety disorders (qandadhd, n.d.).

As children with ADHD are often thought to be disruptive and hyperactive, there are methods to mitigate symptoms. The most common of these are medication such as methylphenidate, a drug that works by stimulating parts of the brain that are

underactive in people with ADHD, helping increase concentration and decrease hyperactivity (YoungMinds, n.d.). However, there are other types of therapy and even methods such as mindfulness can also be employed to help people learn to treat ADHD children and adults (Bachmann et al., 2022). Treatment through medication or other forms is very important due to ADHD being an uncurable condition and helping to alleviate symptoms make the condition easier to cope with.

A method that has been researched to help mitigate the symptoms of ADHD is through the use of music. This can be done through means such as listening to music, music therapy, playing an instrument or even composition. Music has been viewed as a useful method to aid people with ADHD as in most cases it is well structured and has a clear line to follow, helping people with ADHD learn to predict and react to patterns through that music, helping improve their sense of structure (Rodgers, 2022). In addition, as some symptoms of ADHD are linked to lower levels of dopamine, listening to music is said to increase dopamine levels in the brain, which will put people with ADHD in better moods and also motivate them to be more productive (AmeliaVirtualCare, n.d.). The reason behind the lack of dopamine in those with ADHD is because they often have a defunct DRD_4 receptor, making it difficult for the neurons to respond to the dopamine neurotransmitter, which plays an important role in attentiveness (Blum et al, 2008)

This essay will determine to what extent does listening to music lessen the symptoms of ADHD. Listening to music will encompass listening to music whilst working and listening to music during music therapy. The main symptoms of ADHD that are being observed for improvement are attention and hyperactivity to see if

listening to music can help to lessen these symptoms and if so, how significant this effect is. The improvements will be measured in both observational and self-reported scenarios, with most of the participants in these studies having official ADHD diagnoses.

Research from studies done by Madjar et al (2020) provides evidence towards music being a useful tool in increasing the concentration and performance of students in areas such as comprehension whilst Jackson (2003) shows its utility and perceived effectiveness as a music therapy treatment. Montello and Coons (1998) show that listening to music helps concentration and is more effective than actively engaging with it. However, Pelham et al (2011) shows that music can actually being more of a distraction than a tool to help those with ADHD concentrate and Vorster (2020) explains that whilst listening to music can have some effect, it is lesser than just taking medication.

Main Body

A study by Madjar et al (2020) tests the effects listening to music has on reading comprehension of preadolescents both with and without ADHD. Reading comprehension can be difficult for those with ADHD because a lack of attentiveness often causes interest in reading it to be lost, or lack of care in reading it, often resulting in smaller details being missed, losing concentration or losing where they were on the page (Low, 2022). In the group with ADHD, there were 20 participants and in the group without there were 25 participants with the mean age of participants being around 10 years old. Both groups were set a series of reading tasks to complete, however these were done under 4 different conditions. The first condition was without any background music, the second was with calm background music without lyrics, the third was calm background music with lyrics and the final was very rhythmic music with lyrics. Under all conditions where music was used, there was a significant increase in the performance of the participants doing the comprehension who had ADHD, resulting in a higher mark under music conditions than without. By contrast, the performance of those without ADHD in the comprehension who listened to music decreased, ending up with a lower mark than when they did the comprehension without music. This shows that listening to music can decrease symptoms of ADHD, helping to increase attentiveness and performance in the reading comprehension task.

However, whilst this seems to provide some good evidence in favour of music improving attentive skills in those with ADHD, further testing is needed to see if this correlates to other skills such as problem solving in subjects like maths. The improvements to attentive and retentive skills may only be visible and observable in

situations where reading is needed. To test to see if this improvement is noticeable across all subjects, tests that include problem solving should be conducted in similar ways as this one was.

However, this research was done in 2020, which is very recently. This means that this is representative of children of our age and done under a setting where both those with and without ADHD have a greater idea of what it is to be ADHD. Also, as music has been vastly different over the ages, the study being conducted ensures that music used was recent, meaning it has high temporal validity and in turn high external validity showing that the ways listening to music lessons symptoms of ADHD is valid in a modern-day environment.

Another study that shows the effectiveness of listening to music to decrease the symptoms of ADHD is a study conducted by Jackson (2003). The aim of the study was to determine what music therapy treatment early elementary school children with ADHD were receiving, the effectiveness of this and how it compared to other treatments of ADHD. The study was carried out by randomly selecting 500 board certified music therapists from the American Music Therapy Association and sending off a survey to them that was specifically created for this study. The survey asked questions like whether they were currently working with early elementary school children with ADHD, what methods they used and how often they used them, the general perceived effectiveness of the therapy as treatment for the ADHD, ranging from a rating between 1 (not very effective) and 5 (extremely effective) and what other treatments were the patient also undergoing for their ADHD. The findings of the study showed that the most used methods of music therapy were listening to

music and movement and that the perceived effectiveness of these treatments were also high, coming in at a mean of about 4.1. This showed that music therapy is perceived to be effective and a good way of treating children with ADHD.

However, it was also shown that a large number of these children were also undergoing other treatments for ADHD too, the largest of these being medication, in which 87% of the children the therapists were working with were said to be taking. Medication is another method used to treat ADHD and one of the most commonly used. As the majority of children were taking this, it's possible that the medication could have influenced the results to an extent, making it seem as if the music therapy treatment was more effective than it actually was.

In addition, this was a survey filled out by the therapists. When answering a survey, it's possible that people may answer in a way that makes them look more desirable, leading to social desirability bias impacting the results of the experiment and therefore lowering the internal validity of the experiment. Even if they do not believe that the treatment has been particularly effective, they will want to make it look like they've been making progress and having success in treating the ADHD just to make themselves look good. This shows that this may not be a good measure of if listening to music in music therapy is an effective way to lessen symptoms of ADHD. In addition, how different people respond to treatment may vary, so whilst a therapist may believe that treatment is effective, the recipient child may not view it as being effective. This shows that data collected in this study may be limited as it has not considered the effectiveness of the treatment from the child's point of view. A

possible way to improve this then would be to conduct interviews with the child to measure how effective they believed the treatment was.

A study by Montello and Coons (1998) goes further than just testing the effectiveness of listening to music lessening symptoms of ADHD, but rather delves into how listening to music in a passive way may be more effective than engaging with it actively. The aim of the study was to compare and contrast the behavioural effects of more active and rhythmic group therapy vs passive listening based group music therapy. Active music therapy involves participants actively engaging with the music and the rhythm, whilst passive music therapy involves listening to it and noticing patterns within it (Mcpherson et al, 2019). 16 preadolescents ranging from 11-14 years old, with 14 of them being male and 2 being female were enrolled in a special educational program in New York. Whilst not every participant had ADHD, this was among one of the learning difficulties some participants, with others having symptoms very similar to that of ADHD such as short attention span, hyperactivity among other symptoms too. The true intentions of the study were kept hidden from the participants as sessions were disguised as music lessons. Participants were split into 3 groups and each group would receive a 45-minute session every week for 24 weeks. Group A had 6 participants and underwent active music therapy for the first 12 weeks before switching to passive for the last 12. Group B had 4 participants who received passive music therapy for 12 weeks and active for the last 12 and Group C with 6 participants had active music therapy for the entire duration. The teachers of the participants would fill out a form to determine how much of an improvement they perceived was happening as a result of the music therapy. This was filled out before the treatment began, after the first 6 months and at the very end. The findings of the

study showed a greater improvement in symptoms when treated using passive group music therapy as opposed to active group music therapy, showing that music therapy as a treatment is better at decreasing symptoms of ADHD when participants are not engaging with the music actively, rather listening to it passively and carrying out other tasks.

Most participants in this group were made up of boys and there is evidence to suggest that ADHD symptoms, specifically those which present themselves are hyperactivity and impulsivity are less severe and noticeable in girls (Healthline, n.d.), with symptoms such as inattentiveness and anxiety being among the more common symptoms in females (Connolly, n.d.). Whilst this study provides evidence towards passive music therapy being useful in lessening symptoms of ADHD in boys, there is less evidence that it will help females with ADHD cope with the different symptoms they often encounter. Using a more balanced group of males and females or testing the experiment again in a group of only girls may be a better idea to discover if passive music has a similar improvement on girls with ADHD and to what extent the improvement is in comparison to boys.

However, by disguising the experiment as a music lesson it also decreases the likelihood of demand characteristics impacting the results of the experiment as the participants are unaware they are taking part in a research study making it unlikely t they would act in a certain way to make a specific outcome more likely. The study also took place over a span of 6 months, meaning the results of this experiment were not a one off.

By contrast, some studies provide evidence that listening to music may not be entirely beneficial for those suffering with ADHD and can have limited and inconsistent effects, sometimes even acting as a larger distraction in some circumstances. A study conducted by Pelham et al (2011) tests the possible effects that music can have as distractors as well as impacts on behaviour and performance on boys with ADHD. 41 boys with ADHD and 21 without were chosen to participate in the experiment. They were enlisted in an 8-week course called the Summer Treatment Programme. The first 2 weeks served as a way to settle and familiarise students and teachers, but during the third week the distractor variables were introduced. There were 3 variables used. No distractor, music as a distractor and video as a distractor. These variables were randomly allocated to students but it was made sure that each student received at least one of each treatment per week. In addition, since only 24 days of the course remained, it was ensured that each participant underwent each variable only 8 times each. To measure how well they concentrated, a point system was used, so anytime they disobeyed a classroom rule such as not paying attention to a teacher, not respecting other and not working quietly, they lost a point. The results of the experiment showed that under all circumstances, the ADHD participants were much less focused on their work and task and lost more points overall with the no distraction group being overall better, providing evidence that listening to music may worsen symptoms of ADHD and make it more difficult to keep focus instead of lessoning symptoms and increasing focus.

A downside to how this study was conducted was that using a point-based system is very vague way to measure behaviour. It also cannot measure the degree of poor

behaviour of a student rather just the frequency of it. In addition, it's possible different teachers might interpret bad behaviour and lack of concentration in different ways, making one teacher more prone to detract points from students than another different teacher. However, despite this, there still seemed to be a correlation in results between more frequent bad behaviour and listening to music, so despite a point-based system being a vague way to measure behaviour, there still seemed to be some evidence of a correlation between these two factors. In addition it would be have better to collect data on the behaviour of participants prior to the experiment in order to directly compare and contrast how the participants act before and after the experiment to further increase the reliability of the results and ensure that listening to music impacts those with ADHD negatively.

Another study by Vorster (2020) aimed to determine if different musical conditions could affect and improve the concentration of those diagnosed with ADHD. A group of 11 volunteers made up of 8 females and 3 males with a group mean age of 21.85 took part in a 4-day experiment. Each day a different treatment was used. The first was binaural beats, a perception of sound created in your mind when you listen to two tones at a different frequency in different ears, creating a third tone known as a binaural beat. (Sleep Foundation, n.d.) The second was classical music, the third music of the participants' preference and finally ADHD medication. In order to test concentration, 4 different behavioural tasks were used, Stroop tasks, sustained attention tasks, working memory tasks and perception tasks. Additionally, an infrared spectroscope was used to observe the prefrontal cortex and monitor changes in the oxy- and deoxy- levels as well as haemoglobin, which would be indicative of how active and focused on the task participants were. The results of the experiment

showed that whilst classical and music of preference did provide some improvement to the concentration of participants, binaural beats proved to be a larger distraction, and by far the most effective method seemed to be the ADHD tablets. This shows that whilst listening to music can improve symptoms of ADHD, it is not the most efficient or even effective way to control these symptoms.

However, the study mentions that this effect was only significant during the sustained attention task and there were only minimal differences observed during the other 3 conditions. This could mean that only in some circumstances listening to binaural beats and music is less effective than medication. As being prescribed medication may be too expensive for some adults or they may not want to take it, listening to music or binaural beats may still be an effective way to help those with ADHD sustain their attention and continue to work for longer. Furthermore, this study did provide some evidence that they are as effective as each other under 3 out of the 4 total conditions.

Another critical thinking point is that the variable of “music of preference” is a very vague due to there being so many different genres of music. It is entirely possible that some of the participants’ music of preference could have been classical, meaning that the second variable was being repeated again. It would be better to replace music of preference with some high energy, fast rhythmic music like rock music, giving a stark contrast to the classical music and providing a more concrete variable and gain a clearer idea as to the effect listening to music has on the symptoms of ADHD.

Discussion

There are some considerations to take into account with all of these studies. One thing all these studies have not taken into account is non-Western music, as music from continents like Africa and Asia are often very different to music from Europe and America. Eastern music, for example music from India is built upon a melodic framework called a raga, a set of notes used in Indian classical music ("AQA", n.d.). In addition, the music is more improvisational and layered than Western music. Researching the effects, strengths and weaknesses of music that comes from different cultures could be useful as their different structural format and sound may impact someone with ADHD in a completely different way than Western music would.

Another point of discussion is age. Every study used children or young adults as their participants. This is significant as there is research to suggest that as people grow older, symptoms of ADHD can develop and change, presenting themselves differently in adults and sometimes leading to difficulty spotting ADHD in adults and common misdiagnosis too. (Cherry, 2022) For example, it is common in adults for hyperactivity to decrease as they grow older, but likely that problems concentrating and focusing on tasks will stay as well as other potential symptoms such as restlessness, impatience and difficulty dealing with stress. (NHS, n.d.). Because symptoms can develop and change in the lead up to adulthood, believing that just because something helps control the symptoms of ADHD in children must mean it also helps adults isn't a fair claim to make. Further research should be done into the effect of listening to music on adult symptoms of ADHD, possibly even with a

comparison between the effectiveness of this as a treatment between adults and children diagnosed with ADHD.

These experiments are rather standardized though, with a clear aim and procedure making them easy for others to replicate. Finding participants with and without ADHD would not be too difficult too. This gives these experiments a higher reliability making them easily testable in order to see if similar results are obtained through the same method.

Conclusion

This essay discussed the extent that listening to music lessened the symptoms of ADHD. The studies used in this essay showed evidence that music can improve symptoms such as concentration and hyperactivity. However, there are a lot of symptoms of ADHD, many of which can vary depending on gender and age, meaning that most of these studies did not test older adults to see if these methods had similar results. In addition, most of the participants in this experiment were males and since evidence shows some symptoms such as hyperactivity are often lesser in females with ADHD, treatment through listening to music may not be as effective in females as it is in males. Some of the studies in this experiment provided evidence that listening to music was detrimental to the concentration of those with ADHD. However, often a point-based system was used making it difficult to properly monitor and more subject to bias and manipulation. Another study showed that whilst music had some effect, it was lesser than just taking tablets. In addition, none of these studies explored music genres from other cultures, mainly focusing on Western music. Despite this, research and analysis of these studies still show that listening to music has some effect and use in lessening symptoms of ADHD in those who suffer from it. This has some useful implications as often listening to music is free and easy and could be used more commonly in therapy treatments for those struggling with their ADHD.

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